JK BOSE Class 12 Annual Examination

JK BOSE Class 12 Mathematics 6006-B (2022)

Math · Full marks 100 · 150 minutes · 29 questions

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Instructions

- Total number of questions: 29.
- Section A: Multiple Choice Questions, 1 mark each (Q1-Q4).
- Section B: Very Short Answer Type Questions, 2 marks each (Q5-Q12).
- Section C: Short Answer Type Questions, 4 marks each (Q13-Q23).
- Section D: Long Answer Type Questions with internal Or choice, 6 marks each (Q24-Q29).

Section-A (MCQ) — 1 mark each

Q1. The relation R in the set $\{1, 2, 3\}$ given by $R = \{(x, y) : x > y,\ x, y \in A\}$ is: [1 mark · Easy]

(a) Reflexive
(b) Symmetric
(c) Transitive
(d) None of these

Q2. The principal value of $\cos^{-1}\left(-\frac{1}{2}\right)$ is: [1 mark · Easy]

(a) $\frac{\pi}{3}$
(b) $\frac{\pi}{6}$
(c) $\frac{2\pi}{3}$
(d) $\frac{\pi}{4}$

Q3. If A, B are symmetric matrices of same order, then $AB - BA$ is a: [1 mark · Medium]

(a) Symmetric matrix
(b) Skew symmetric matrix
(c) Zero matrix
(d) Identity matrix

Q4. The direction cosines of a unit vector along z-axis are: [1 mark · Easy]

(a) $(1, 0, 0)$
(b) $(0, 1, 0)$
(c) $(0, 0, 1)$
(d) $(1, 1, 1)$

Section-B (Very Short Answer) — 2 marks each

Q5. Find x and y if: $2\begin{bmatrix} 1 & 3 \\ 0 & x \end{bmatrix} + \begin{bmatrix} y & 0 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 5 & 6 \\ 1 & 8 \end{bmatrix}$ [2 marks · Easy]

Q6. Evaluate: $\int (4e^{3x} + 1)\, dx$ [2 marks · Easy]

Q7. Verify that the function $y = x^2 + 2x + C$ is the solution of the differential equation $\frac{dy}{dx} - 2x - 2 = 0$. [2 marks · Easy]

Q8. Find the rate of change of the area of a circle with respect to its radius r when $r = 4$ cm. [2 marks · Easy]

Q9. Find the angle between the vectors $\hat{i} - 2\hat{j} + 3\hat{k}$ and $3\hat{i} - 2\hat{j} + \hat{k}$. [2 marks · Easy]

Q10. Solve the following L.P.P. graphically: Maximise $Z = 3x + 4y$ subject to the constraints $x + y \leq 4, x \geq 0, y \geq 0$. [2 marks · Easy]

Q11. If $P(A) = \frac{6}{11}$, $P(B) = \frac{5}{11}$ and $P(A \cup B) = \frac{7}{11}$, find $P(A|B)$. [2 marks · Easy]

Q12. Given two independent events A and B such that $P(A) = 0.3$, $P(B) = 0.6$. Find $P(A \text{ or } B)$. [2 marks · Easy]

Section-C (Short Answer) — 4 marks each

Q13. Given $f : \mathbb{R} \to \mathbb{R}$ given by $f(x) = 4x + 3$. Show that f is invertible. Find the inverse of f. [4 marks · Medium]

Q14. Write the following function in the simplest form: $\tan^{-1}\left[\frac{\cos x - \sin x}{\cos x + \sin x}\right]$, $0 < x < \pi$. [4 marks · Medium]

Q15. Express the matrix $A = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}$ as the sum of symmetric and a skew-symmetric matrix. [4 marks · Medium]

Q16. Find the relationship between a and b so that the function f defined by $f(x) = \begin{cases} ax + 1 & \text{if } x \leq 3 \\ bx + 3 & \text{if } x > 3 \end{cases}$ is continuous at $x = 3$. [4 marks · Medium]

Q17. Find the intervals in which the function f given by $f(x) = 4x^3 - 6x^2 - 72x + 30$ is (a) strictly increasing, (b) strictly decreasing. [4 marks · Medium]

Q18. Find the equations of the tangent and normal to the given curve at the indicated point: $y = x^4 - 6x^3 + 13x^2 - 10x + 5$ at $(0, 5)$. [4 marks · Medium]

Q19. Find the general solution of the differential equation: $\frac{dy}{dx} = \frac{1+y^2}{1+x^2}$ [4 marks · Medium]

Q20. Find the area of the region bounded by the ellipse: $\frac{x^2}{16} + \frac{y^2}{9} = 1$ [4 marks · Medium]

Q21. Find $|\vec{a} \times \vec{b}|$ if $\vec{a} = \hat{i} - 7\hat{j} + 7\hat{k}$ and $\vec{b} = 3\hat{i} - 2\hat{j} + 2\hat{k}$. [4 marks · Medium]

Q22. If $\vec{a} = 5\hat{i} - \hat{j} - 3\hat{k}$ and $\vec{b} = \hat{i} + 3\hat{j} - 5\hat{k}$, then show that $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$ are perpendicular. [4 marks · Easy]

Q23. Solve the following problem graphically: Minimise and maximise $Z = 3x + 9y$ subject to the linear constraints $x + 3y \leq 60$, $x \leq y$, $x + y \geq 10$, $x \geq 0$, $y \geq 0$. [4 marks · Hard]

Section-D (Long Answer, internal Or choice) — 6 marks each

Q24. By using properties of determinants prove that: $\begin{vmatrix} a & a+b & a+b+c \\ 2a & 3a+2b & 4a+3b+2c \\ 3a & 6a+3b & 10a+6b+3c \end{vmatrix} = a^3$ [6 marks · Hard]

Or

If $A = \begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$, then verify that $A(\text{adj } A) = |A|I$. Also find A^{-1}.

Q25. Find $\frac{dy}{dx}$ of the function $x^y + y^x = 1$. [6 marks · Hard]

Or

If $y = 3\cos(\log x) + 4\sin(\log x)$, show that $x^2 y_2 + x y_1 + y = 0$.

Q26. Integrate the rational fraction: $\frac{2x-3}{(x^2-1)(2x+3)}$ [6 marks · Hard]

Or

Using the properties of definite integrals evaluate: $\int_{-5}^{5} |x+2|\, dx$

Q27. Find the general solution of the differential equation: $x\frac{dy}{dx} + 2y = x^2 \log x$ [6 marks · Hard]

Or

Show that the differential equation $(x^2 - y^2)\, dx + 2xy\, dy = 0$ is homogeneous and solve it.

Q28. Find the equation of the plane through the intersection of the planes $3x - y + 2z - 4 = 0$ and $x + y + z - 2 = 0$ and the point $(2, 2, 1)$. [6 marks · Hard]

Or

Find the angle between the line $\frac{x+1}{2} = \frac{y}{3} = \frac{z-3}{6}$ and the plane $10x + 2y - 11z = 3$.

Q29. Find the probability distribution of number of doublets in three throws of a pair of dice. [6 marks · Hard]

Or

Two balls are drawn at random with replacement from a box containing 10 black balls and 8 red balls. Find the probability that: (a) both balls are red, (b) first ball is black and second is red, (c) one of them is black and other is red.

Solutions

Generated during digitization — the source paper ships without a key. Review before student-facing use.

Q1.

Answer: (C) Transitive

- **List the pairs:** On $A = \{1, 2, 3\}$, $R = \{(2, 1), (3, 1), (3, 2)\}$.
- **Check reflexive and symmetric:** $(1, 1) \notin R$ so not reflexive; $(2, 1) \in R$ but $(1, 2) \notin R$ so not symmetric.
- **Check transitive:** $x > y$ and $y > z$ always give $x > z$; e.g. $(3, 2), (2, 1) \in R$ and $(3, 1) \in R$. Hence transitive.

The 'greater than' relation is transitive but neither reflexive nor symmetric, since $x > x$ is never true and $x > y$ never implies $y > x$.

Q2.

Answer: (C) $\frac{2\pi}{3}$

- **Recall the principal range:** The principal value branch of $\cos^{-1}$ is $[0, \pi]$.
- **Solve in that range:** $\cos\theta = -\frac{1}{2}$ with $\theta \in [0, \pi]$ gives $\theta = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$.

Within $[0, \pi]$ the only angle whose cosine is $-\frac{1}{2}$ is $\frac{2\pi}{3}$.

Q3.

Answer: (B) Skew symmetric matrix

- **Take the transpose:** $(AB - BA)^T = (AB)^T - (BA)^T = B^T A^T - A^T B^T$.
- **Use symmetry of A and B:** $A^T = A$, $B^T = B$, so this equals $BA - AB = -(AB - BA)$.
- **Conclude:** A matrix equal to the negative of its own transpose is skew-symmetric.

Transposing $AB - BA$ and using $A^T = A$, $B^T = B$ flips its sign, which is exactly the definition of a skew-symmetric matrix.

Q4.

Answer: (C) $(0, 0, 1)$

- **Write the unit vector:** Along the z-axis the unit vector is $\hat{k} = 0\hat{i} + 0\hat{j} + 1\hat{k}$.
- **Read off direction cosines:** For a unit vector the direction cosines are its components: $(0, 0, 1)$.

A unit vector's components are its direction cosines, and $\hat{k}$ has components $(0, 0, 1)$.

Q5.

Answer: $x = 3$, $y = 3$

- **Compute the left side:** $\begin{bmatrix} 2 & 6 \\ 0 & 2x \end{bmatrix} + \begin{bmatrix} y & 0 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 2+y & 6 \\ 1 & 2x+2 \end{bmatrix}$.
- **Equate entries:** $2 + y = 5 \Rightarrow y = 3$; $2x + 2 = 8 \Rightarrow x = 3$.

Matrix equality means every corresponding entry is equal, giving two simple linear equations.

Q6.

Answer: $\frac{4}{3}e^{3x} + x + C$

- **Integrate term by term:** $\int 4e^{3x} dx = \frac{4}{3}e^{3x}$ and $\int 1\, dx = x$.
- **Add the constant:** Result: $\frac{4}{3}e^{3x} + x + C$.

$\int e^{ax}dx = \frac{e^{ax}}{a}$, applied with $a = 3$, plus the integral of the constant 1.

Q7.

Answer: Verified: substituting $\frac{dy}{dx} = 2x + 2$ makes the equation identically zero.

- **Differentiate:** $y = x^2 + 2x + C \Rightarrow \frac{dy}{dx} = 2x + 2$.
- **Substitute:** $(2x + 2) - 2x - 2 = 0$, so the equation is satisfied for every C.

A function solves a differential equation when substituting it (and its derivatives) makes the equation an identity.

Q8.

Answer: 8π cm^2/cm

- **Set up:** $A = \pi r^2$, so $\frac{dA}{dr} = 2\pi r$.
- **Evaluate at r = 4:** $\left.\frac{dA}{dr}\right|_{r=4} = 2\pi(4) = 8\pi$ cm^2/cm.

The rate of change of area with respect to radius is the derivative $2\pi r$, evaluated at the given radius.

Q9.

Answer: $\theta = \cos^{-1}\left(\frac{5}{7}\right)$

- **Dot product:** $\vec{a} \cdot \vec{b} = 3 + 4 + 3 = 10$.
- **Magnitudes:** $|\vec{a}| = |\vec{b}| = \sqrt{14}$.
- **Apply the formula:** $\cos\theta = \frac{10}{14} = \frac{5}{7}$, so $\theta = \cos^{-1}\frac{5}{7}$.

The angle between two vectors comes from $\cos\theta = \frac{\vec{a}\cdot\vec{b}}{|\vec{a}||\vec{b}|}$.

Q10.

Answer: $Z_{\text{max}} = 16$ at $(0, 4)$

- **Identify the feasible region:** Triangle with corner points $(0, 0)$, $(4, 0)$, $(0, 4)$ in the first quadrant under $x + y \leq 4$.
- **Evaluate Z at corners:** $Z(0, 0) = 0$, $Z(4, 0) = 12$, $Z(0, 4) = 16$.
- **Pick the maximum:** Maximum $Z = 16$ at $(0, 4)$.

For a linear objective over a convex polygon, the optimum occurs at a corner point; the largest corner value here is 16.

Q11.

Answer: $P(A|B) = \frac{4}{5}$

- **Find the intersection:** $P(A \cap B) = P(A) + P(B) - P(A \cup B) = \frac{6}{11} + \frac{5}{11} - \frac{7}{11} = \frac{4}{11}$.
- **Apply conditional probability:** $P(A|B) = \frac{P(A\cap B)}{P(B)} = \frac{4/11}{5/11} = \frac{4}{5}$.

Conditional probability divides the joint probability by the probability of the conditioning event.

Q12.

Answer: $P(A \cup B) = 0.72$

- **Use independence:** $P(A \cap B) = P(A)P(B) = 0.3 \times 0.6 = 0.18$.
- **Addition rule:** $P(A \cup B) = 0.3 + 0.6 - 0.18 = 0.72$.

For independent events the joint probability is the product, and the addition rule then gives the union.

Q13.

Answer: f is one-one and onto, hence invertible; $f^{-1}(y) = \frac{y-3}{4}$

- **Show one-one:** $f(x_1) = f(x_2) \Rightarrow 4x_1 + 3 = 4x_2 + 3 \Rightarrow x_1 = x_2$.
- **Show onto:** For any $y \in \mathbb{R}$, $x = \frac{y-3}{4}$ is real and $f(x) = y$.
- **Write the inverse:** Solving $y = 4x + 3$ for x: $f^{-1}(y) = \frac{y-3}{4}$.

A bijective function is invertible, and solving $y = f(x)$ for x yields the inverse formula.

Q14.

Answer: $\frac{\pi}{4} - x$

- **Divide by cos x:** $\frac{\cos x - \sin x}{\cos x + \sin x} = \frac{1-\tan x}{1+\tan x}$.
- **Recognise the tangent identity:** $\frac{1-\tan x}{1+\tan x} = \tan\left(\frac{\pi}{4} - x\right)$.
- **Simplify:** $\tan^{-1}\left[\tan\left(\frac{\pi}{4} - x\right)\right] = \frac{\pi}{4} - x$.

The expression matches the tangent subtraction formula $\tan(\frac{\pi}{4} - x)$, so the inverse tangent collapses to $\frac{\pi}{4} - x$.

Q15.

Answer: $A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T)$ with symmetric part $\begin{bmatrix} 2 & -3/2 & -3/2 \\ -3/2 & 3 & 1 \\ -3/2 & 1 & -3 \end{bmatrix}$ and skew part $\begin{bmatrix} 0 & -1/2 & -5/2 \\ 1/2 & 0 & 3 \\ 5/2 & -3 & 0 \end{bmatrix}$

- **Write the transpose:** $A^T = \begin{bmatrix} 2 & -1 & 1 \\ -2 & 3 & -2 \\ -4 & 4 & -3 \end{bmatrix}$.
- **Symmetric part:** $P = \frac{1}{2}(A + A^T) = \begin{bmatrix} 2 & -3/2 & -3/2 \\ -3/2 & 3 & 1 \\ -3/2 & 1 & -3 \end{bmatrix}$, and $P^T = P$.
- **Skew-symmetric part:** $Q = \frac{1}{2}(A - A^T) = \begin{bmatrix} 0 & -1/2 & -5/2 \\ 1/2 & 0 & 3 \\ 5/2 & -3 & 0 \end{bmatrix}$, and $Q^T = -Q$.
- **Verify:** $P + Q = A$. ✓

Every square matrix decomposes uniquely as the sum of its symmetric part $\frac{1}{2}(A + A^T)$ and skew-symmetric part $\frac{1}{2}(A - A^T)$.

Q16.

Answer: $3a - 3b = 2$, i.e. $a = b + \frac{2}{3}$

- **Left and right limits:** $\lim_{x\to 3^-} f = 3a + 1$; $\lim_{x\to 3^+} f = 3b + 3$; $f(3) = 3a + 1$.
- **Impose continuity:** $3a + 1 = 3b + 3 \Rightarrow 3a - 3b = 2 \Rightarrow a = b + \frac{2}{3}$.

Continuity at a point requires the left limit, right limit, and function value to coincide, which forces the linear relation between a and b.

Q17.

Answer: Strictly increasing on $(-\infty, -2) \cup (3, \infty)$; strictly decreasing on $(-2, 3)$

- **Differentiate:** $f'(x) = 12x^2 - 12x - 72 = 12(x - 3)(x + 2)$.
- **Find critical points:** $f'(x) = 0$ at $x = -2$ and $x = 3$.
- **Sign analysis:** $f' > 0$ for $x < -2$ or $x > 3$ (increasing); $f' < 0$ for $-2 < x < 3$ (decreasing).

The sign of the factored derivative $12(x-3)(x+2)$ on each interval determined by the critical points decides where f increases or decreases.

Q18.

Answer: Tangent: $10x + y = 5$; Normal: $x - 10y + 50 = 0$

- **Slope at the point:** $\frac{dy}{dx} = 4x^3 - 18x^2 + 26x - 10$; at $x = 0$ the slope is -10.
- **Tangent line:** $y - 5 = -10(x - 0) \Rightarrow 10x + y = 5$.
- **Normal line:** Normal slope $= \frac{1}{10}$: $y - 5 = \frac{1}{10}x \Rightarrow x - 10y + 50 = 0$.

The tangent uses the derivative's value as slope at the point, and the normal is perpendicular, so its slope is the negative reciprocal.

Q19.

Answer: $\tan^{-1} y = \tan^{-1} x + C$

- **Separate variables:** $\frac{dy}{1+y^2} = \frac{dx}{1+x^2}$.
- **Integrate both sides:** $\tan^{-1} y = \tan^{-1} x + C$.

The equation is separable, and both sides integrate to inverse tangents.

Q20.

Answer: 12π square units

- **Set up by symmetry:** $a = 4$, $b = 3$; area $= 4\int_0^4 \frac{3}{4}\sqrt{16 - x^2}\, dx$.
- **Integrate:** $3\left[\frac{x\sqrt{16-x^2}}{2} + 8\sin^{-1}\frac{x}{4}\right]_0^4 = 3 \cdot 8 \cdot \frac{\pi}{2}$.
- **Conclude:** Area $= 12\pi$ sq units (consistent with πab).

Integrating the upper-half curve over one quadrant and using four-fold symmetry gives the standard ellipse area πab.

Q21.

Answer: $|\vec{a} \times \vec{b}| = 19\sqrt{2}$

- **Cross product determinant:** $\vec{a} \times \vec{b} = \hat{i}(-14 + 14) - \hat{j}(2 - 21) + \hat{k}(-2 + 21) = 19\hat{j} + 19\hat{k}$.
- **Magnitude:** $|\vec{a} \times \vec{b}| = \sqrt{0 + 19^2 + 19^2} = 19\sqrt{2}$.

Expanding the 3×3 determinant gives the cross product vector, whose magnitude is $19\sqrt{2}$.

Q22.

Answer: $(\vec{a} + \vec{b}) \cdot (\vec{a} - \vec{b}) = 0$, hence perpendicular

- **Form the sum and difference:** $\vec{a} + \vec{b} = 6\hat{i} + 2\hat{j} - 8\hat{k}$; $\vec{a} - \vec{b} = 4\hat{i} - 4\hat{j} + 2\hat{k}$.
- **Dot product:** $(6)(4) + (2)(-4) + (-8)(2) = 24 - 8 - 16 = 0$.
- **Conclude:** Zero dot product means the vectors are perpendicular. ✓

Two nonzero vectors are perpendicular exactly when their dot product is zero.

Q23.

Answer: $Z_{\text{min}} = 60$ at $(5, 5)$; $Z_{\text{max}} = 180$ at every point of the segment joining $(15, 15)$ and $(0, 20)$

- **Find corner points:** Feasible-region corners: $(0, 10)$, $(5, 5)$, $(15, 15)$, $(0, 20)$.
- **Evaluate Z:** $Z(0, 10) = 90$, $Z(5, 5) = 60$, $Z(15, 15) = 180$, $Z(0, 20) = 180$.
- **Interpret:** Minimum 60 at $(5, 5)$. Maximum 180 occurs at two corners, hence at every point of the segment between $(15, 15)$ and $(0, 20)$.

Corner-point evaluation gives the extremes; equal maxima at two adjacent corners means the whole joining edge is optimal.

Q24.

Answer: Proved: the determinant reduces to $a \cdot a^2 = a^3$ after row operations

- **Row operations:** $R_2 \to R_2 - 2R_1$, $R_3 \to R_3 - 3R_1$ give rows $(0, a, 2a+b)$ and $(0, 3a, 7a+3b)$.
- **Expand along column 1:** $= a\,[a(7a+3b) - 3a(2a+b)]$.
- **Simplify:** $= a(7a^2 + 3ab - 6a^2 - 3ab) = a \cdot a^2 = a^3$. ■

Subtracting multiples of the first row creates zeros in the first column, collapsing the determinant to a 2×2 that simplifies to a^2.

Or — If $A = \begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$, then verify that $A(\text{adj } A) = |A|I$. Also find A^{-1}.

Answer: $|A| = 1$; $A^{-1} = \text{adj } A = \begin{bmatrix} 7 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$

- **Determinant:** $|A| = 1(16-9) - 3(4-3) + 3(3-4) = 7 - 3 - 3 = 1$.
- **Adjugate:** Cofactor matrix transposed: adj $A = \begin{bmatrix} 7 & -3 & -3 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$.
- **Verify and invert:** $A(\text{adj } A) = I = |A|I$ ✓; since $|A| = 1$, $A^{-1} = \text{adj } A$.

The identity $A(\text{adj } A) = |A|I$ holds for every square matrix, and with $|A| = 1$ the adjugate itself is the inverse.

Q25.

Answer: $\frac{dy}{dx} = -\frac{y\,x^{y-1} + y^x \log y}{x^y \log x + x\,y^{x-1}}$

- **Differentiate x^y:** $\frac{d}{dx}x^y = x^y\left(\frac{y}{x} + \log x \frac{dy}{dx}\right)$ (logarithmic differentiation).
- **Differentiate y^x:** $\frac{d}{dx}y^x = y^x\left(\log y + \frac{x}{y}\frac{dy}{dx}\right)$.
- **Solve for dy/dx:** Sum equals 0; collecting $\frac{dy}{dx}$ terms gives $\frac{dy}{dx} = -\frac{yx^{y-1} + y^x \log y}{x^y \log x + xy^{x-1}}$.

Both terms are variable-to-variable powers, so each is differentiated logarithmically and the resulting linear equation in $\frac{dy}{dx}$ is solved.

Or — If $y = 3\cos(\log x) + 4\sin(\log x)$, show that $x^2y_2 + xy_1 + y = 0$.

Answer: Shown: differentiating twice and substituting gives $x^2y_2 + xy_1 + y = 0$

- **First derivative:** $xy_1 = -3\sin(\log x) + 4\cos(\log x)$.
- **Differentiate again:** $y_1 + xy_2 = \frac{-3\cos(\log x) - 4\sin(\log x)}{x} = -\frac{y}{x}$.
- **Multiply by x:** $xy_1 + x^2y_2 = -y \Rightarrow x^2y_2 + xy_1 + y = 0$. ■

Differentiating the first-derivative relation once more reproduces $-y/x$, which rearranges to the required second-order equation.

Q26.

Answer: $-\frac{1}{10}\log|x-1| + \frac{5}{2}\log|x+1| - \frac{12}{5}\log|2x+3| + C$

- **Factor and decompose:** $\frac{2x-3}{(x-1)(x+1)(2x+3)} = \frac{A}{x-1} + \frac{B}{x+1} + \frac{C}{2x+3}$.
- **Find constants:** $A = -\frac{1}{10}$ (at $x = 1$), $B = \frac{5}{2}$ (at $x = -1$), $C = -\frac{24}{5}$ (at $x = -\frac{3}{2}$).
- **Integrate:** $-\frac{1}{10}\log|x-1| + \frac{5}{2}\log|x+1| + \frac{C}{2}\log|2x+3| = \ldots - \frac{12}{5}\log|2x+3| + C_1$.

Partial fractions split the rational function into three simple terms, each integrating to a logarithm; note $\int \frac{dx}{2x+3} = \frac{1}{2}\log|2x+3|$.

Or — Using the properties of definite integrals evaluate: $\int_{-5}^{5} |x+2|\,dx$

Answer: 29

- **Split at the kink:** $|x+2| = -(x+2)$ for $x < -2$ and $(x+2)$ for $x \geq -2$.
- **Evaluate both pieces:** $\int_{-5}^{-2} -(x+2)dx = \frac{9}{2}$; $\int_{-2}^{5}(x+2)dx = \frac{49}{2}$.
- **Add:** $\frac{9}{2} + \frac{49}{2} = 29$.

The absolute value changes definition at $x = -2$, so the integral is split there and each triangular piece is evaluated.

Q27.

Answer: $y = \frac{x^2}{4}\log x - \frac{x^2}{16} + \frac{C}{x^2}$

- **Standard linear form:** $\frac{dy}{dx} + \frac{2}{x}y = x\log x$.
- **Integrating factor:** $e^{\int \frac{2}{x}dx} = x^2$, so $\frac{d}{dx}(yx^2) = x^3\log x$.
- **Integrate by parts:** $yx^2 = \frac{x^4}{4}\log x - \frac{x^4}{16} + C \Rightarrow y = \frac{x^2}{4}\log x - \frac{x^2}{16} + \frac{C}{x^2}$.

This is a first-order linear ODE; multiplying by the integrating factor x^2 makes the left side an exact derivative, and the right side integrates by parts.

Or — Show that the differential equation $(x^2 - y^2)\,dx + 2xy\,dy = 0$ is homogeneous and solve it.

Answer: $x^2 + y^2 = Cx$

- **Show homogeneity:** $\frac{dy}{dx} = \frac{y^2-x^2}{2xy}$ depends only on $\frac{y}{x}$, hence homogeneous.
- **Substitute y = vx:** $v + x\frac{dv}{dx} = \frac{v^2-1}{2v} \Rightarrow \frac{2v\,dv}{1+v^2} = -\frac{dx}{x}$.
- **Integrate:** $\log(1+v^2) = -\log x + \log C \Rightarrow x(1+\frac{y^2}{x^2}) = C \Rightarrow x^2 + y^2 = Cx$.

Writing the equation as a function of y/x proves homogeneity, and the substitution $y = vx$ separates the variables.

Q28.

Answer: $7x - 5y + 4z - 8 = 0$

- **Family of planes:** $(3x - y + 2z - 4) + \lambda(x + y + z - 2) = 0$.
- **Use the point:** At $(2, 2, 1)$: $2 + 3\lambda = 0 \Rightarrow \lambda = -\frac{2}{3}$.
- **Substitute back:** $\frac{7}{3}x - \frac{5}{3}y + \frac{4}{3}z - \frac{8}{3} = 0 \Rightarrow 7x - 5y + 4z - 8 = 0$ (check: $14 - 10 + 4 - 8 = 0$ ✓).

Any plane through the line of intersection belongs to the one-parameter family; the given point fixes the parameter.

Or — Find the angle between the line $\frac{x+1}{2} = \frac{y}{3} = \frac{z-3}{6}$ and the plane $10x + 2y - 11z = 3$.

Answer: $\theta = \sin^{-1}\left(\frac{8}{21}\right)$

- **Direction and normal:** Line direction $\vec{b} = (2, 3, 6)$; plane normal $\vec{n} = (10, 2, -11)$.
- **Apply the formula:** $\sin\theta = \frac{|\vec{b}\cdot\vec{n}|}{|\vec{b}||\vec{n}|} = \frac{|20+6-66|}{7\times 15} = \frac{40}{105} = \frac{8}{21}$.

The angle between a line and a plane is the complement of the angle with the normal, giving the sine formula used here.

Q29.

Answer: $P(X=0) = \frac{125}{216}$, $P(X=1) = \frac{75}{216}$, $P(X=2) = \frac{15}{216}$, $P(X=3) = \frac{1}{216}$

- **Probability of a doublet:** $P(\text{doublet}) = \frac{6}{36} = \frac{1}{6}$ per throw; $X \sim B(3, \frac{1}{6})$.
- **Binomial formula:** $P(X=r) = \binom{3}{r}\left(\frac{1}{6}\right)^r\left(\frac{5}{6}\right)^{3-r}$.
- **Tabulate:** $\frac{125}{216}, \frac{75}{216}, \frac{15}{216}, \frac{1}{216}$ for $r = 0, 1, 2, 3$ (sums to 1 ✓).

Each throw is an independent trial with doublet probability $\frac{1}{6}$, so the count of doublets follows a binomial distribution.

Or — Two balls are drawn at random with replacement from a box containing 10 black balls and 8 red balls. Find the probability that: (a) both balls are red, (b) first ball is black and second is red, (c) one of them is black and other is red.

Answer: (a) $\frac{16}{81}$; (b) $\frac{20}{81}$; (c) $\frac{40}{81}$

- **Per-draw probabilities:** With replacement: $P(\text{red}) = \frac{8}{18} = \frac{4}{9}$, $P(\text{black}) = \frac{10}{18} = \frac{5}{9}$, draws independent.
- **Multiply:** (a) $\left(\frac{4}{9}\right)^2 = \frac{16}{81}$; (b) $\frac{5}{9} \cdot \frac{4}{9} = \frac{20}{81}$.
- **Either order:** (c) black-red or red-black: $2 \cdot \frac{20}{81} = \frac{40}{81}$.

Replacement keeps the draws independent and identically distributed, so each event's probability is a simple product.